

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SECURITY INCIDENT MAPPING SYSTEM (SIM)

Submitted By

Name: Babatimilehin Dolapo Victoria

Matric Number: 241201213

Department: Computer Science

Course Code: SEN 202

Course Title: Introduction to Software Engineering II

Institution: Adekunle Ajasin University, Akungba-Akoko, Ondo State

Submission Date: July 2026

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CHAPTER ONE

1.0 INTRODUCTION

1.1 Background of the Study

Security is a fundamental requirement for maintaining a peaceful and productive learning environment in higher institutions. Universities are expected to provide an atmosphere where students, lecturers, staff, and visitors can carry out academic and social activities without fear of theft, assault, harassment, vandalism, or other security threats. As universities continue to expand in population and infrastructure, ensuring adequate security becomes increasingly challenging. Traditional methods of monitoring and reporting incidents are often insufficient to address the dynamic security needs of modern campuses.

In many universities, security incidents are reported manually through verbal complaints, handwritten reports, or direct visits to security offices. While these methods have been used for many years, they are often slow, inefficient, and difficult to manage. Manual reporting may lead to delays in response, poor record keeping, and difficulties in identifying crime patterns across different locations on campus. In some cases, students may choose not to report incidents because the reporting process is stressful or inconvenient.

The rapid growth of information and communication technology has transformed the way organizations manage information and deliver services. Web-based applications have become essential tools for improving communication, increasing accessibility, and organizing information efficiently. Educational institutions are increasingly adopting digital solutions to enhance administrative processes, academic activities, and student welfare.

Applying these technologies to campus security can significantly improve the reporting and management of security incidents.

The Security Incident Mapping System (SIM) is proposed as a web-based platform designed to simplify the reporting of security-related incidents within the university environment. The system enables users to submit reports through an online form while also providing access to safety announcements and a visual crime map that highlights locations where incidents have occurred. By making security information more accessible, the system promotes awareness, encourages prompt reporting, and contributes to a safer campus community.

Although this project is implemented as a static front-end application using HTML, CSS, and JavaScript, it serves as a functional prototype that demonstrates the interface and core features of a modern campus security reporting system. Future versions of the application can incorporate databases, authentication systems, geographic information systems, and administrative dashboards to create a fully operational security management solution.

Furthermore, the development of the Security Incident Mapping System demonstrates the practical application of software engineering principles, including requirements analysis, system planning, interface design, and software documentation. It highlights the importance of understanding user requirements before software development and provides a structured foundation upon which future enhancements can be built.

1.2 Problem Statement

Security-related incidents remain a major concern in many tertiary institutions. Cases of theft, harassment, assault, vandalism, and suspicious activities can affect students' academic performance, reduce confidence in campus security, and create an unsafe learning environment. Despite the presence of security personnel, many institutions still rely on manual methods for receiving and processing incident reports.

Manual reporting systems present several challenges. Victims or witnesses are often required to visit security offices physically before submitting complaints, which may discourage timely reporting. Reports may be misplaced, incomplete, or difficult to retrieve when needed for future reference. In addition, security personnel may find it difficult to analyze manually stored reports to identify crime trends or areas requiring increased surveillance.

Another challenge is the lack of a centralized platform through which members of the university community can receive timely security information. Students are often unaware of recent incidents occurring around campus, making it difficult for them to avoid potentially unsafe locations. The absence of accessible safety updates reduces awareness and limits proactive measures that could improve campus security.

There is therefore a need for a user-friendly digital platform that simplifies incident reporting while improving communication between students, staff, and campus security personnel. The proposed Security Incident Mapping System addresses these challenges by providing an organized, accessible, and efficient platform for reporting incidents, viewing safety announcements, and promoting security awareness within the university community.

1.3 Aim of the Study

The primary aim of this project is to develop a Security Incident Mapping System (SIM) that provides a simple, user-friendly, and web-based platform for reporting security incidents within the university environment. The system is intended to improve campus safety by enabling users to submit incident reports conveniently while providing access to relevant security information and updates.

In addition, the project aims to demonstrate the practical application of software engineering principles in the analysis, design, and specification of a web-based information system using modern web technologies such as HTML, CSS, and JavaScript.

1.4 Objectives of the Study

The specific objectives of this project are to:

Design a simple and responsive web application for reporting security incidents.

Provide users with an easy-to-use interface for submitting incident reports.

Display campus safety announcements and security updates.

Present a visual crime map showing sample incident locations.

Improve awareness of security issues within the university community.

Encourage timely reporting of security incidents.

Demonstrate the application of software engineering principles in system development.

Develop a prototype using HTML, CSS, and JavaScript without relying on backend technologies.

1.5 Scope of the Study

The scope of this project is limited to the design and development of a static web-based Security Incident Mapping System (SIM). The system focuses on providing an interactive interface through which users can explore the various features of a campus security application.

The website consists of several interconnected pages including the Home page, Login page, Registration page, Report Incident page, Crime Map page, Safety Updates page, and About page. Each page has been designed to perform a specific function while maintaining a consistent layout and navigation structure throughout the application.

The Home page introduces users to the system and provides quick access to important features. The Login and Registration pages demonstrate how users would access the platform in a real-world implementation. The Report Incident page provides a form through which users can submit details of security incidents. The Crime Map page presents a visual representation of reported incidents using sample markers, while the Safety Updates page displays important security announcements. The About page provides background information regarding the purpose and objectives of the system.

This project focuses solely on the front-end aspect of software development. It does not include server-side programming, database management, user authentication, cloud storage, real-time communication, or live mapping services. The website serves as a prototype intended to demonstrate the proposed functionality and user interface of a complete Security Incident Mapping System.

1.6 Significance of the Study

The Security Incident Mapping System is significant because it demonstrates how information technology can be used to improve campus safety through efficient communication and digital reporting. As educational institutions continue to adopt technology-driven solutions, the development of systems that enhance security becomes increasingly important.

The proposed system benefits several groups within the university community.

Students

Students are the primary beneficiaries of the system. They can easily access security information, learn about recent incidents, and understand how security issues can be reported through a digital platform. The availability of safety updates also helps students make informed decisions regarding their movements within the campus.

University Security Personnel

The proposed system provides a structured method of collecting security reports. Instead of relying entirely on manual documentation, the platform demonstrates how incident information can be organized digitally, making future management more efficient.

University Management

The project illustrates how technology can support institutional decision-making by providing a centralized platform for reporting and monitoring security-related issues. Such a system can contribute to improved planning and resource allocation for campus security.

Software Engineering Students

The project serves as an academic example of software requirements analysis and web application design. It demonstrates how software engineering concepts such as requirements gathering, documentation, interface design, and system planning can be applied in solving real-world problems.

Researchers and Future Developers

The report provides a foundation upon which future researchers and developers can build more advanced versions of the Security Incident Mapping System by incorporating databases, authentication systems, and real-time mapping technologies.

1.7 Limitations of the Study

Although the Security Incident Mapping System demonstrates the essential features of a campus security reporting platform, it has certain limitations due to its academic nature and implementation approach.

The system is developed as a static front-end website and therefore does not support dynamic data processing. Information entered into forms is not stored permanently because no database has been integrated into the application.

The Login and Registration pages are designed only to demonstrate the user interface and basic validation. They do not perform actual user authentication or account management.

Similarly, the Crime Map page presents sample incident locations using placeholder markers rather than real geographic data. The map does not communicate with external mapping services such as Google Maps or OpenStreetMap.

The system also lacks an administrative dashboard through which reports can be reviewed, verified, or managed. Features such as email notifications, SMS alerts, report tracking, and emergency response services are outside the scope of this project.

Despite these limitations, the system successfully demonstrates the functional requirements and interface design expected of a Security Incident Mapping System and provides a strong foundation for future development.

1.8 Definition of Terms

Software Requirements Specification (SRS): A formal document that describes the functional and non-functional requirements of a software system and serves as a guide during software development.

Security Incident: Any event or activity that threatens the safety of individuals, property, or the university environment. Examples include theft, assault, harassment, vandalism, and suspicious activities.

Incident Report: A record containing information submitted by a user describing a security-related event that has occurred within the campus environment.

Crime Map: A graphical representation of locations where security incidents have been reported. It helps users identify areas that may require caution.

User: Any student, staff member, visitor, or resident who accesses the Security Incident Mapping System to obtain information or submit reports.

Administrator: An authorized individual responsible for monitoring the system, managing reports, and publishing safety announcements in a fully implemented version of the application.

Frontend: The visual component of a web application with which users directly interact. It includes web pages, forms, navigation menus, buttons, and other interface elements.

Static Website: A website whose content is fixed and does not rely on server-side processing or database interactions. All information is presented directly through HTML, CSS, and JavaScript.

Campus Security: The collection of policies, personnel, and technological measures implemented to ensure the safety of students, staff, visitors, and university property.

CHAPTER TWO

SOFTWARE REQUIREMENTS SPECIFICATION

2.1 Introduction

A Software Requirements Specification (SRS) is a document that defines the requirements and expected behavior of a software system before development begins. It serves as a guide for developers, designers, testers, and other stakeholders by describing what the system should do and the conditions under which it should operate.

This chapter presents the requirements for the proposed Security Incident Mapping System (SIM). It outlines the purpose of the system, its scope, major functions, intended users,

operating environment, and the functional and non-functional requirements necessary for the successful implementation of the application.

2.2 Purpose of the Proposed System

The purpose of the Security Incident Mapping System is to provide a simple platform through which members of the university community can report security incidents and access important safety information. The system is intended to improve awareness of campus security issues while demonstrating the use of web technologies in solving real-world problems.

The proposed system also serves as a prototype that illustrates how software engineering principles can be applied in designing a practical information system.

2.3 Scope of the System

The Security Incident Mapping System is designed as a static web application developed using HTML, CSS, and JavaScript.

The system includes the following features:

Home page

User registration

User login

Incident reporting form

Crime map

Safety updates

About page

The project focuses on interface design and user interaction. Features such as database management, user authentication, and real-time data processing are not included in this implementation.

2.4 Overall Description

The Security Incident Mapping System is designed to provide users with quick access to security-related information through a simple and responsive website.

Users can navigate through different pages to register, log in, submit an incident report, view sample crime locations, and read campus safety announcements. The website is

designed with a consistent layout to ensure ease of navigation and a positive user experience.

Although the current version is static, the structure has been designed to support future integration with backend technologies and databases.

2.5 Product Perspective

The Security Incident Mapping System is an independent web application developed to improve security awareness within a university environment. It acts as a centralized platform where users can obtain security information and understand how incidents would be reported in a fully implemented system.

The application is intended for educational purposes and demonstrates the expected functionality of a campus security management system. It is designed to be scalable, allowing additional features to be incorporated in future versions.

2.6 Product Functions

The major functions of the system include:

Displaying information about the Security Incident Mapping System.

Allowing users to register and log into the platform.

Providing an online incident reporting form.

Displaying a sample crime map showing reported incidents.

Publishing campus safety announcements.

Providing information about the purpose of the system.

These functions collectively provide users with an overview of how a modern campus security platform operates.

2.7 User Characteristics

The proposed system is intended for the following categories of users:

Students

Students can access safety information, register, log in, and submit incident reports.

University Staff

Staff members can use the system to stay informed about security issues and report incidents occurring within the campus.

Visitors

Visitors can view general information about the system and read available safety updates.

Security Personnel

Security officers can use the information provided by the system to understand reported incidents and monitor security awareness activities in future implementations.

2.8 Operating Environment

The Security Incident Mapping System is designed to operate on modern web browsers, including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

The website is compatible with desktop computers, laptops, tablets, and smartphones. It requires only a web browser to function and does not require additional software installation.

The application is developed using HTML5 for page structure, CSS3 for styling, and JavaScript for client-side interactions.

2.9 Functional Requirements

Functional requirements describe the services and operations that the Security Incident Mapping System is expected to perform. These requirements define how users interact with the system and the features that should be available.

FR1: Home Page

The system shall provide a homepage that introduces users to the Security Incident Mapping System. The homepage shall display a welcome message, a brief description of the system, links to important pages, recent security alerts, and emergency contact information.

FR2: User Registration

The system shall provide a registration page where new users can enter their personal information, including their full name, student identification number, email address, phone number, and password. Basic input validation shall ensure that all required fields are completed before submission.

FR3: User Login

The system shall provide a login page that allows registered users to enter their email address and password before accessing the platform. JavaScript shall be used to perform simple client-side validation of user inputs.

FR4: Incident Reporting

The system shall provide an online incident report form where users can submit details of security incidents. The form shall include fields such as incident type, location, date, description, and an option to upload supporting evidence. Upon submission, the system shall display a confirmation message indicating that the report has been successfully submitted.

FR5: Crime Map

The system shall include a crime map page displaying sample incident locations using markers. The map serves as a visual representation of areas where security incidents have occurred and is intended for demonstration purposes only.

FR6: Safety Updates

The system shall display safety announcements and security awareness messages. These updates provide users with useful information regarding campus safety and recommended precautionary measures.

FR7: About Page

The system shall include an About page explaining the purpose of the Security Incident Mapping System, its objectives, and the benefits it provides to the university community.

2.10 Non-Functional Requirements

Non-functional requirements describe the quality attributes of the proposed system.

Performance

The website should load quickly and respond smoothly when users navigate between pages.

Usability

The interface should be simple, organized, and easy to understand. Users should be able to locate information without difficulty.

Reliability

The system should function correctly under normal operating conditions without broken links or interface errors.

Maintainability

The website should be developed using clean and well-structured code to simplify future updates and modifications.

Portability

The application should function on different operating systems and web browsers without significant changes.

Accessibility

The website should use readable fonts, clear navigation, and appropriate colour contrast to ensure accessibility for different categories of users.

2.11 External Interface Requirements

User Interface

The user interface is designed with simplicity and ease of use in mind. The website features a consistent navigation menu, responsive layout, readable typography, clearly labelled buttons, and organized content sections. The colour scheme uses warm golden yellow, deep brown, and off-white to create a professional and welcoming appearance.

Software Interface

The application is developed using HTML5, CSS3, and JavaScript. It requires only a standard web browser and does not depend on any external libraries or software frameworks.

Hardware Interface

The system can be accessed using desktop computers, laptops, tablets, and smartphones with internet browser support.

2.12 Assumptions and Constraints

The development of the Security Incident Mapping System is based on several assumptions.

It is assumed that users have access to a device capable of running a modern web browser. It is also assumed that users possess basic computer literacy and can navigate through the website without extensive guidance.

Since the project is implemented as a static front-end application, several constraints exist. The system does not include a backend server, database, or real-time authentication. Incident reports are not permanently stored, and user accounts are not managed beyond

basic interface demonstrations. Similarly, the crime map uses sample markers instead of live geographic information.

These constraints are acceptable because the primary objective of the project is to demonstrate software requirements specification and front-end system design rather than full-scale implementation.

2.13 Future Enhancements

Although the current version of the Security Incident Mapping System is a static prototype, it provides a solid foundation for future development.

Future versions of the system may include database integration for storing user accounts and incident reports permanently. A secure authentication system could also be implemented to enable users to log in securely and monitor the status of submitted reports.

The crime map could be integrated with mapping technologies such as Google Maps or OpenStreetMap to display real-time incident locations. Additional features such as image uploads, email notifications, SMS alerts, emergency SOS services, and live security announcements could further improve the effectiveness of the application.

An administrator dashboard could also be developed to enable security personnel to manage reports, update safety announcements, monitor user activities, and generate security statistics for decision-making.

These enhancements would transform the prototype into a comprehensive campus security management system capable of supporting effective communication and improving safety across the university community.

2.14 Conclusion

The Security Incident Mapping System (SIM) was proposed to provide a simple and effective approach to reporting and monitoring security incidents within a university environment. As educational institutions continue to adopt digital technologies, there is an increasing need for systems that promote safety, improve communication, and encourage timely reporting of security-related issues.

This Software Requirements Specification has presented a detailed description of the proposed system, including its objectives, scope, overall design, functional requirements, non-functional requirements, operating environment, and implementation constraints. These requirements establish a clear foundation for the development of the application and ensure that the system is designed to meet the needs of its intended users.

The prototype developed for this project focuses on the front-end implementation using HTML, CSS, and JavaScript. Although it does not include backend processing or database integration, it successfully demonstrates the structure and functionality of a modern security reporting platform. Users are able to navigate through the various pages, understand the reporting process, view safety information, and experience the overall workflow of the proposed system.

Furthermore, the project demonstrates the importance of software requirements analysis in software engineering. Clearly identifying system requirements before implementation helps reduce development errors, improves planning, and ensures that the final product satisfies user expectations. This project therefore serves as both an academic exercise and a practical example of how software engineering principles can be applied to address real-world challenges.

In conclusion, the Security Incident Mapping System provides a valuable framework for improving campus security through technology. With future enhancements such as database integration, real-time mapping, secure authentication, and administrative management tools, the system has the potential to evolve into a fully functional campus security platform capable of supporting safer learning environments and more efficient incident management. Although this implementation is limited to a static front-end prototype, it successfully demonstrates the feasibility, usefulness, and future potential of the proposed system.